2024 Ph H2 Q5

Section: Our Dynamic Universe

Topic: The Expanding Universe (Doppler & Evidence)

Summary of Question

A class demonstration models wavefronts from a ‘star’ using students. Initially the ‘star’ is stationary; students (wavefronts) leave the ‘star’ every 3.5 s and walk towards the observer at 1.5 m s^{-1}. The demo is repeated with the ‘star’ moving relative to the observer, and you must decide towards/away with justification. Finally, connect the dark night sky (Olbers’ paradox) and another observation to the expanding Universe.

(a) Wavelength of the light (from the model)

In the model, wavelength = distance between successive wavefronts = (speed of students) × (time between wavefronts).

$\lambda = 1.5 \times 3.5 = 5.25$ m.

Final answer: 5.25 m

(b) Is the ‘star’ moving towards or away? Justify

Answer: The ‘star’ is moving away from the observer.

Justification: At the observer, the spacing between wavefronts is larger (fewer per second arrive) than when the source was stationary → lower observed frequency → redshift → source receding.

(c)(i) Olbers’ (dark sky) paradox and expansion

If the Universe were static, infinite and eternal, the night sky would be bright (light from infinitely many stars would fill every line of sight).

But the sky is dark: the Universe has a finite age and is expanding, so much starlight has not yet arrived and much radiation has been redshifted beyond the visible band.

Final point: The dark sky supports a finite-age, expanding Universe.

(c)(ii) One other piece of evidence for expansion

Hubble’s Law: recession velocity of galaxies is proportional to their distance ($v \propto d$).

Acceptable alternatives: cosmic microwave background radiation or primordial light-element abundances.

Revision Tips

In Doppler contexts, be explicit: more/fewer wavefronts per second ↔ higher/lower observed frequency.

For ‘evidence’, give a specific named item (e.g., Hubble’s Law), not just the generic term ‘redshift’.

State final answers clearly and concisely with units (e.g., 5.25 m).